



AuthenticOne — Requirements

Requirements are organized around **three perspectives**: what the trader does, what the admin manages, and what the system does automatically.

Key Decisions (Resolved)

- **Trading is fully automated.** Once the Decision Agent validates a trade, the platform sends the order straight to the user's connected broker — no manual "click to trade" step.
 - **Two AI agents, two jobs:**
 - **Decision Agent** (inside Signal Generation) — reads market state/structure and decides: valid trade or not, and if valid, buy or sell.
 - **Monitoring Agent** (inside Risk) — owns everything after entry: position sizing, entry style (full / partial / scaled), stop loss, take profit, and live trade management (breakeven moves, target adjustments, scaling).
 - **One broker** connection per user — no multi-broker for now.
 - Platform doesn't distinguish real vs. paper accounts — it just sends orders to whatever broker account is connected.
 - **Strategy is fully AI-decided** — no user-selected strategies.
 - **Signals/analytics run for every supported asset simultaneously** — the platform controls the supported-asset list; a user's watchlist just controls what they focus on, not what gets computed.
 - **Backtesting is out of scope** for now.
 - **Admin has no special trading privileges** — to trade, an admin uses a normal account like anyone else.
 - **Performance is tracked in two separate views:** account performance (trader-facing) and strategy performance (admin/system-facing, since strategies belong to the platform).
-

1. Non-Functional Requirements

- **Security** — Protect accounts, broker credentials, and trading data from unauthorized access.
 - **Performance** — Fast enough for user-facing actions and low-latency for real-time data/signals/execution.
 - **Reliability** — Handle broker/API failures, DB failures, lost connections, failed orders, and restarts gracefully.
 - **Observability** — Enough logging/monitoring to reconstruct what happened in any trading workflow.
 - **Maintainability** — New assets, features, or agent logic can be added without breaking unrelated parts.
-

2. User / Trader Requirements

Account

- Register, log in, log out
- Recover/reset password
- View and edit profile

Subscription

- View plans
- Subscribe / cancel
- View current subscription status
- **Free:** create account, watchlists, add assets, view a limited/minimal version of analytics per asset
- **Subscribed:** full access — full analytics, AI signals, automated execution

Watchlists

- Create, rename, delete, organize watchlists
- Add/remove assets from the platform's supported-asset list
- View asset info

Asset Analytics

- View analytics per asset (price, volume, volatility, indicators, trend, regime, AI insights, etc.)
- View historical and real-time analytics
- Choose which analytics to display
- Depth of analytics shown depends on subscription tier

AI Signals

- View signals produced by the Decision Agent (asset, direction, confidence, features, status)
- View how the Monitoring Agent is managing an open trade (size, stop, target, adjustments)
- View signal history and outcomes
- No strategy selection — the AI decides how it trades

Broker Integration

- Connect **one** broker account
- View broker account info, positions, orders, and executions
- No distinction made between real-money and paper accounts — orders just go to whatever is connected

Trading

- Trading is automated: valid signals are executed without a manual confirm step
- Set risk guardrails (see Risk section below)
- Monitor active trades in real time
- Modify or close positions manually if needed

Performance & History

- View **account performance** (P&L, win rate, drawdown, Sharpe ratio, etc.) — this is the trader's own results
- View and filter trading history

- (Strategy-level performance is an admin/system view, not a trader view — see Admin section)

Notifications

- Get notified on: new signal, order submitted/executed/rejected, position closed, risk limit reached, broker disconnected, subscription events, system events
-

3. Admin Requirements

User Management

- View, search, filter users
- View user details
- Enable/disable accounts
- Manage roles
- View and manage user subscriptions

Subscription Management

- Create/modify/enable/disable plans
- Configure what each tier unlocks

Platform Monitoring

- View system status (data feeds, DB, processing, signals, broker connections, execution engine)
- View system errors and logs
- Monitor data ingestion, signal generation, order execution

Trading Oversight

- View platform-wide trading activity, exposure, active positions, order activity
- View execution failures and risk events
- Configure global risk controls
- Trigger a **global trading halt** in emergencies

- Admin has **no personal trading privileges** — must use a standard user account to trade like anyone else

Strategy Performance

- View strategy-level performance (win rate, expectancy, drawdown, etc. — aggregated across all users), since strategies are owned by the platform, not individual traders
-

4. System Requirements (automatic behavior)

Market Data

- Retrieve real-time and historical data from external providers, for the full platform-supported asset list
- Store and normalize data
- Handle data-source failures

Feature Processing

- Turn raw data into mathematical features
- Support both real-time and batch/historical generation
- Keep real-time and historical calculations consistent

Signal Generation — Decision Agent

- Evaluate market state/structure for every supported asset
- Decide: valid trade or not
- If valid: output direction (buy/sell)
- Record each decision and link it to the features behind it
- Track signal outcomes over time

Risk Management — Monitoring Agent

- Determine position size and entry style (full / partial / scaled) for a validated trade
- Set stop loss and take profit

- Manage the trade live: move stop to breakeven, adjust targets, scale in/out as conditions evolve
- Enforce user-set guardrails:
 - Max consecutive losing trades per day
 - Max consecutive losing trades per week
 - Max loss per day
 - Max loss per week
- Enforce platform-level (admin-configured) global risk controls

Execution

- Convert approved decisions into orders
- Submit orders to the user's connected broker
- Track order status and fills
- Handle rejected orders and broker failures
- Reconcile internal state with actual broker state

Cross-Cutting

- **End-to-end traceability:** Market Data → Features → Signal → Risk Decision → Order → Execution → Position → Outcome
- **Idempotent order processing** — no duplicate execution on retries
- **State consistency** between internal records and broker
- **Audit trail** for user, admin, trading, risk, and system actions
- **Backtesting is not in scope** for now